Pandas Cheatsheet — Essential Functions (A4)

Data Loading & Inspection	
`pd.read_csv("file.csv")`	Load CSV
`pd.read_json("data.json")`	Load JSON
`pd.read_excel("file.xlsx")`	Load Excel
`df.info()`	Schema + nulls
`df.head(10)`	Preview rows
`df.tail(5)`	Last rows
`df.sample(5)`	Random sample
`df.shape`	Rows/columns
`df.dtypes`	Column types
`df.columns`	Column names
`df.describe()`	Summary stats

Selecting & Filtering	
`df['col']`	Select column
`df[['a','b']]`	Select multiple columns
`df.iloc[0:5]`	Select by position
`df.loc[df['x']>0]`	Select by label/condition
`df[df['age'] > 30]`	Boolean filter
`df[(df['age']>18) & (df['country']=='US')]`	Multi-condition
`df.query("country=='US'")`	SQL-style filter
`df[df['age'].between(20,30)]`	Range filter
`df[df['id'].isin([1,2,3])]`	Is-in filter

Type Handling	
`df['id'].astype('Int64')`	Cast type
`df['id'].astype(str)`	To string
`pd.to_datetime(df['ts'])`	To datetime
`df['ts'].dt.date`	Extract date
`df['ts'].dt.year`	Extract year
`df['ts'].dt.month`	Extract month
`df['ts'].dt.dayofweek`	Day of week

Cleaning & Transforming	
`df.dropna(subset=...) / how='all' / inplace=True`	Drop missing rows
`df.fillna(0) / fillna(..., inplace=True)`	Fill missing
`df.ffill()`	Forward fill
`df.drop_duplicates(...) / duplicated()`	Dedupe / flag dupes
`df.columns.difference([...])`	Exclude columns
`df.reset_index(drop=True) / ..., inplace=True`	Reset index
`df.rename(columns={...}) / ..., inplace=True`	Rename cols
`df.drop(columns=[...]) / ..., inplace=True`	Drop cols
`df.replace({...})`	Replace values
`df.assign(new_col=...)`	Add column
`np.where(cond, a, b)`	Conditional column

Aggregation & Grouping	
`df.groupby('cat')['sales'].sum()`	Single col group
`df.groupby(['cat','region'])['sales'].sum()`	Multi-col group
`df.groupby(['cat','region']).agg({...})`	Multi-col + multi-agg
`df.groupby('cat').mean()`	Group mean
`df.groupby('id').agg({'a':'sum','b':'count'})`	Multi-agg
`df.groupby('cat').size()`	Group size
`df['id'].count()`	Count
`df['user_id'].nunique()`	Unique count
`df['country'].value_counts()`	Frequency
`df.sort_values('sales', ascending=False)`	Sort
`df.sort_values('sales', inplace=True)`	Sort in-place
`df.nlargest(5, 'sales')`	Top N rows

Joining & Combining	
`pd.merge(a, b, on='id', how='left')`	Left join
`pd.merge(a, b, on='id', how='right')`	Right join
`pd.merge(a, b, on='id', how='inner')`	Inner join
`pd.merge(a, b, on='id', how='outer')`	Outer join
`pd.merge(a, b, how='cross')`	Cross join
`df1.join(df2, how='left')`	Index join
`pd.concat([df1, df2])`	Stack DataFrames
`pd.concat([df1, df2], axis=1)`	Concat columns

Advanced — Window & Cumulative	
`df['ma7'] = df['sales'].rolling(7).mean()`	Rolling mean
`df['lag'] = df['x'].shift(1)`	Lag column
`df['r'] = df.groupby('cat')['sales'].rank()`	Rank
`df.groupby('cat').cumcount()`	Cumcount per group
`df['x'].cumsum()`	Cumulative sum
`df.groupby('cat')['x'].cumsum()`	Cumsum per group
`df['x'].cummax()`	Cumulative max
`df['x'].cummin()`	Cumulative min
`df['x'].pct_change()`	Pct change
`df['x'].diff(1)`	Diff (lag diff)

Advanced — Reshape & Strings	
`df.pivot_table(values='sales', index='cat', columns='region')`	Pivot
`df.melt(id_vars=['id'])`	Wide to long
`df.explode('col')`	Explode list col
`df['x2'] = df['x'].apply(lambda x: x*2)`	Apply
`df['email'].str.lower()`	String lower
`df['col'].str.contains('x')`	String contains
`df['col'].str.split(',')`	String split
`df.loc[df['age']<0, 'age'] = 0`	Update rows

80+ ops • 3-column layout • A4